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CONTINUAL LEARNING FOR OBJECT CLASSIFICATION USING GAUSSIAN MODELING (PRELIMINARY TITLE)

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CONTINUAL LEARNING FOR OBJECT CLASSIFICATION USING GAUSSIAN MODELING (PRELIMINARY TITLE)

Esta investigación aborda el desafío del aprendizaje continuo para la clasificación y detección de objetos en tiempo real. Se centra en la integración de un extractor de características preentrenado de manera no supervisada con modelos Gaussianos independientes para caracterizar las clases de objetos. El principal objetivo de este trabajo es desarrollar y validar un sistema capaz de incorporar nuevas clases de forma incremental sin necesidad de reentrenar el modelo completo, mitigando el olvido catastrófico de las categorías previamente aprendidas y permitiendo la detección automática de clases desconocidas en escenarios de aplicación reales.

Para lograrlo, el estudio evalúa la efectividad de representaciones Gaussianas separables y no separables, analiza su compacidad y eficiencia muestral, y explora su extensión hacia arquitecturas de detección de objetos reemplazando las cabezas de clasificación tradicionales por modelos Gaussianos. Además, la investigación examina el uso de autoencoders como extractores de características adaptados al aprendizaje continuo, y analiza la viabilidad de emplear redes RBF con unidades ocultas Gaussianas como alternativa a los modelos Gaussianos.

El enfoque propuesto busca combinar la independencia de los modelos Gaussianos con la capacidad de generalización del aprendizaje no supervisado de características, proporcionando una solución escalable y eficiente para entornos dinámicos donde continuamente emergen nuevas clases de objetos.

CONTINUAL LEARNING FOR OBJECT CLASSIFICATION USING GAUSSIAN MODELING (PRELIMINARY TITLE)

This research addresses the challenge of continual learning for real-time object classification and detection. It focuses on integrating an unsupervised pretrained feature extractor with independent Gaussian models to characterize object classes. The main objective of this work is to develop and validate a system capable of incrementally incorporating new classes without retraining the entire model, mitigating catastrophic forgetting of previously learned categories, and enabling the automatic detection of unknown classes in real application scenarios. To achieve this, the study evaluates the effectiveness of separable and non-separable Gaussian representations, analyzes their compactness and sample efficiency, and explores their extension to object detection pipelines by replacing traditional classification heads with Gaussian models. Additionally, the research investigates the use of autoencoders as feature extractors adapted for continual learning, and examines the feasibility of employing RBF networks with Gaussian hidden units as an alternative to Gaussian models. The proposed approach seeks to combine the independence of Gaussian models with the generalization capacity of unsupervised feature learning, providing a scalable and efficient solution for dynamic environments where new object classes continuously emerge.

Abstract

Los modelos convencionales de aprendizaje profundo (DL) suelen ser monolíticos y presentan dificultades para adaptarse cuando se enfrentan a nuevas clases, requiriendo típicamente un reentrenamiento completo con todo el conjunto de datos. Esto limita su aplicabilidad en sistemas del mundo real que deben evolucionar e incorporar información nueva con el tiempo.

Para abordar estos desafíos, este trabajo introduce un enfoque de aprendizaje continuo (CL) que desacopla la extracción de características de la clasificación, combinando un extractor de características preentrenado de manera no supervisada con modelos Gaussianos independientes que capturan la representación de cada clase. Con este enfoque, las representaciones Gaussianas pueden ofrecer un aprendizaje continuo estable y eficiente.

Además, se extiende el método a la detección de objetos en tiempo real reemplazando las cabezas de clasificación estándar por nuestros modelos basados en Gaussianas, y se explora el uso de autoencoders como extractores de características alternativos optimizados para CL. También se aborda el problema crítico de identificar automáticamente cuándo debe introducirse una nueva clase, y se investiga el uso de redes de Funciones de Base Radial (RBF) con núcleos Gaussianos como posibles sustitutos de los modelos de clase.

Los experimentos realizados en el benchmark a gran escala ImageNet y en otros conjuntos de datos como Caltech-256 y CIFAR-100 muestran que nuestro método alcanza un rendimiento comparable al de los enfoques de vanguardia basados en repetición, regularización y prompts, con una caída de precisión inferior al 0.5% incluso cuando se agregan nuevas clases de manera secuencial. Estos resultados demuestran que el enfoque propuesto no solo es efectivo y escalable, sino también más simple de implementar que los métodos existentes. Esta simplicidad facilita un aprendizaje clase a clase fluido, permitiendo una adaptación continua a entornos reales dinámicos y en constante evolución.

Abstract

Conventional deep learning (DL) models are often monolithic and struggle to adapt when exposed to new classes, typically requiring retraining on the entire dataset. This limits their applicability in real-world systems that must evolve and incorporate novel information over time. To address these challenges, this work introduces a continual learning (CL) approach that decouples feature extraction from classification, pairing an unsupervised pretrained feature extractor with independent Gaussian models that capture each class representation. With this approach Gaussian representations can provide stable and efficient CL. We further extend the method to real-time object detection by replacing standard classification heads with our Gaussian-based models and explore autoencoders as an alternative feature extractor optimized for CL. Additionally, we tackle the critical problem of automatically identifying when a new class should be introduced and investigate the use of Radial Basis Function (RBF) networks with Gaussian kernels as potential substitutes for the class models. Experiments on the large-scale ImageNet benchmark and other datasets like Caltech-256 and CIFAR-100 reveal that our method reaches performance on par with rehearsal, regularization and prompt-based state-of-the-art approaches , while incurring less than a 0.5% accuracy drop even as new classes are sequentially added. These results demonstrate that the proposed approach is not only effective and scalable, but also simpler to implement compared to the existing approaches. This simplicity facilitates seamless class-by-class learning while enabling continuous adaptation to dynamic and evolving real-world environments.

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Chapter 1

Introduction

1.1 Motivation

State-of-the-art (SOTA) object recognition methods utilize deep learning models that are either trained end-to-end on the target application or initially trained in one application and then fine-tuned for a different target application. In either case, they are monolithic models, meaning that the same single model is used for feature extraction and for classification, and is normally trained on a vast dataset. These models are highly effective when the set of object classes to be recognized is fixed and known at the training/fine-tuning time. However, when new object classes need to be incorporated, the model must be retrained or fine-tuned to accommodate them. This training requires significant computational resources and access to the entire training dataset, and in some cases leads to catastrophic forgetting [1].

In continual or class-incremental object recognition, an algorithm must learn to distinguish between a growing number of object classes [2]. In the context of continual learning, a task refers to a distinct portion of the training experience, usually presented sequentially to the model. Instead of having access to the entire dataset at once, as in standard supervised learning, the model receives data in sequential chunks [3]. Each task generally correspond to a set of new classes to recognize. The main challenge is that when the model is trained task by task, it tends to overwrite previously learned knowledge. Given that current monolithic models are not capable of handling this type of application, an alternative paradigm is required.

This limitation has motivated SOTA methods for object classification to focus on balancing stability and plasticity. Stability refers to a model's ability to preserve previously learned knowledge, while plasticity describes its capacity to adapt and incorporate new tasks [4]. However, traditional approaches rarely optimize both aspects simultaneously: when stability dominates, the model struggles to integrate new knowledge; when plasticity dominates, it loses access to past information. This trade-off is compounded by the limited capacity of neural networks to integrate new representations without interfering with old ones, and by the need to continually adjust the network's structure to match the increasing number of object classes.

One solution to this problem is to divide the monolithic models into two distinct components: one responsible for feature extraction and the other for object classification. These components must be trained separately, at different times. The feature extraction model must be trained when the recognition system is initialized/created, and the obtained features must be general enough to characterize all possible classes of objects to be seen (not just the ones known at training time). The object classification model should be easily updated or trained as new object classes are detected. Ideally, it should be able to incorporate new object classes without modifying the parts of the model responsible for recognizing the already known object classes. This mirrors the functionality of the human visual system, where the structures responsible for the extraction of characteristics (simple and complex cells in cortical areas V1 and V2) undergo adaptation during the initial postnatal weeks/months [5]. However, the ability to learn new categories of objects persists throughout life. Object recognition occurs in high-level cognitive areas, and acquiring knowledge of a new object class can result in a slight reduction in the ability to recognize similar classes of objects over time.

We are inspired by this idea and propose a continual learning object recognition system based on two components. The first component is a feature extraction model built using a pretrained foundation model. The second component is a statistical classifier built using various Gaussian models. Each model (single Gaussian or mixture of Gaussians) characterizes a specific object class independently of the others. Since these Gaussian models can be trained or built at different times, a new Gaussian model is created to characterize a new object class when it needs to be incorporated. For classification, the class whose Gaussian model yields the highest likelihood of the input features is selected. This approach is analogous to a Bayes classifier, in which new classes are incorporated over time.

Gaussian models offer two key advantages for characterizing object classes. First, each object class model operates independently from the others, facilitating the straightforward incorporation of new classes into the model. Second, Gaussian models provide a compact representation of object classes, requiring few samples to construct a model that can be refined as new samples become available.

1.2 Hypothesis

We hypothesize that the use of independent Gaussian models, combined with an unsupervised pretrained feature extractor, enables continual learning that can incorporate new classes in object classification and detection without retraining the entire model. This design mitigates catastrophic forgetting and further allows the automatic detection of previously unknown classes.

1.3 General Objectives

The main objective of this work is to develop and validate a continual learning system for real-time object classification and detection, based on an unsupervised pretrained feature

extractor and independent Gaussian models, with the purpose of enabling the incremental incorporation of new classes, preventing catastrophic forgetting of previously learned classes, and enabling the automatic detection of unknown classes in real application scenarios.

1.4 Specific Objectives

For the purpose of fulfilling the general objective and verifying the hypothesis of this work, the following specific objectives are defined:

- Evaluate Gaussian representations in continual learning. Empirically verify that representing classes using separable and non-separable Gaussians enables stable continual learning without significant degradation in previously learned classes.
- Extend the approach to real-time object detection. Investigate and apply object detection architectures in which the classification head can be modified and replaced by Gaussian models to allow the incremental incorporation of new classes.
- Explore autoencoders as feature extractors. Research, train, and validate an autoencoder that achieves performance comparable to the backbones used so far as feature extractors, but optimized for continual learning environments.
- Implement automatic detection of new classes. Develop a mechanism to automatically identify when an observed object does not belong to any known class, in order to dynamically create new classes in the system.
- Analyze alternatives with RBF networks. Study the feasibility of replacing Gaussian models with RBF networks using Gaussian functions as hidden units, evaluating their performance and viability in incremental learning.

Chapter 2

Theoretical framework

In this chapter we will present the necessary concepts to fully understand the work done. First, we will review general concepts of artificial intelligence (AI), Machine Learning (ML) Image Processing, Object Classification and Object Detection. Second, we will describe the datasets used in this work and finally we will talk about Continual Learning and describe the State-of-the-Art (SOTA) methods.

2.1 Artificial Intelligence

2.1.1 Computer Vision

2.1.2 Machine Learning

2.1.3 Vision Transformers

2.1.4 Object Classification

2.1.5 Object Detection

2.2 Image Datasets

Several image datasets were utilized in the course of this work. A detailed description of each dataset is provided below.

2.2.1 ImageNet

ImageNet [6] is a large-scale visual database widely used in computer vision research. It is organized according to the WordNet hierarchy and provides annotated, high-resolution images across 1,000 object classes, making it a standard benchmark for large-scale image classification. The dataset contains over 1.2 million training images, 50,000 validation images, and 100,000 test images, covering a broad diversity of real-world objects. Its scale, diversity, and hierarchical structure have established ImageNet as a cornerstone resource for evaluating and advancing machine learning models in image recognition and related tasks.

ImageNetR

ImageNet-R (ImageNet-Renditions) [7] is a dataset derived from ImageNet, consisting of images that depict objects through non-photographic styles such as art, cartoons, graffiti, embroidery, origami, paintings, patterns, sculptures, sketches, tattoos, toys, and video game graphics. It covers 200 ImageNet classes and contains approximately 30,000 images. The dataset was constructed by collecting diverse renditions of ImageNet classes and retaining only those samples that standard models, such as ResNet-50, failed to classify correctly. As a result, ImageNet-R provides a challenging benchmark for evaluating the robustness and generalization of visual recognition models beyond natural images.

ImageNet21k

While ImageNet-1K serves as the primary dataset for pretraining deep learning models for computer vision, ImageNet-21K [8] offers a much broader and more diverse coverage of visual concepts. It is an extended version of the ImageNet dataset and contains over 14 million annotated images spanning approximately 21,000 semantic classes derived from the WordNet hierarchy. Although historically less utilized due to its size, complexity, and accessibility challenges, recent advances in preprocessing, hierarchical structuring, and training schemes have demonstrated its significant value for pretraining. Models pretrained on ImageNet-21K have shown improved performance and generalization across a wide range of computer vision tasks, including when transferred to more compact architectures such as mobile-oriented networks.

2.2.2 CIFAR100

CIFAR-100 [9] is an image classification dataset that extends the structure of CIFAR-10 to a more fine-grained setting. It consists of 60,000 color images of size 32×32 pixels, divided into 100 classes with 600 images per class. Each class contains 500 training samples and 100 test samples. The classes are grouped into 20 superclasses, providing both a fine label (the specific class) and a coarse label (the superclass). This hierarchical organization allows for evaluating both fine-grained recognition and broader categorization. The dataset includes a wide variety of semantic groups, such as animals (e.g., aquatic mammals, insects, reptiles),

objects (e.g., household furniture, vehicles), and natural scenes (e.g., forests, mountains, sea), making it a challenging benchmark for image classification and hierarchical learning tasks.

2.2.3 Caltech256

Caltech-256 [10] is a widely used dataset for object recognition research. It contains 30,607 images distributed across 256 object categories plus an additional "clutter" category for background rejection. Each category includes a minimum of 80 images, providing greater balance compared to earlier datasets such as Caltech-101. The images were collected from the web and manually screened to ensure quality, with improvements introduced to avoid artifacts and increase category diversity. Due to its scale and variability, Caltech-256 serves as a challenging benchmark for evaluating classification performance and the generalization ability of visual recognition models.

2.3 Continual Learning

Continual Learning (CL), also known as lifelong learning, constitutes the core focus of this work. It represents a machine learning paradigm designed to handle the progressive acquisition of knowledge from sequences of tasks or data streams over time. In contrast to conventional training approaches, which rely on a fixed dataset and a single training phase, CL emphasizes the ongoing integration of new information while preserving previously learned knowledge. This property is crucial for real-world scenarios where data distributions are non-stationary and environments evolve continuously. The principal challenge lies in mitigating catastrophic forgetting, a phenomenon in which the assimilation of new knowledge leads to the degradation or loss of prior learning [11].

2.3.1 Elasticity and Plasticity

A central concept in continual learning is the balance between elasticity and plasticity. Plasticity refers to the ability of a model to adapt rapidly to new information, ensuring that it can effectively learn novel tasks. Elasticity, in contrast, emphasizes stability by preserving the parameters and representations that are crucial for retaining old knowledge [4]. Achieving an appropriate equilibrium between these two properties is known as the stability–plasticity dilemma: models that are overly plastic may forget past tasks, whereas models that are overly elastic may struggle to integrate new knowledge. Successful continual learning methods are therefore designed to optimize this trade-off.

2.3.2 Main Paradigms

As shown in figure 2.1, two principal paradigms define how continual learning problems are formulated. Task-incremental learning (Task-IL) assumes that the task identity is available

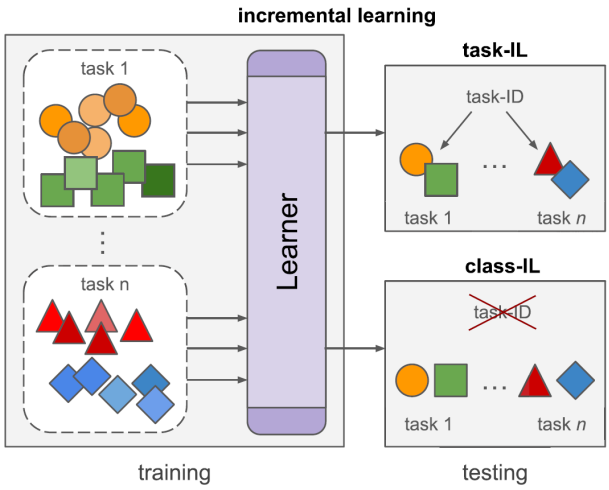


Figure 2.1: Class and Task Incremental Learning [13].

during inference [12]. In this setting, models can rely on task labels to select the appropriate output head, thereby simplifying the classification problem. Class-incremental learning (Class-IL), on the other hand, does not provide the task identity at test time [13]. The model must classify examples among all the classes encountered so far, which makes the problem significantly more challenging. Class-IL is closer to real-world conditions, as systems are rarely informed of which "task" a new input belongs to during deployment.

2.3.3 Mitigation Strategies

In this work, the discussion of continual learning strategies is organized into three main categories. The first category comprises rehearsal-based methods, which mitigate forgetting by storing or generating representative samples from previous tasks and replaying them during training. The second category encompasses regularization-based methods, which constrain parameter updates to preserve knowledge deemed important for earlier tasks, thereby promoting stability without relying on explicit memory storage. Finally, we include representation-based methods, which explore novel ways of encoding and maintaining information to facilitate continual learning. By focusing on these three categories, the analysis encompasses both classical continual learning approaches and recent innovations, such as prompting mechanisms as an alternative to architectural expansion.

2.3.4 Rehearsal Methods

Rehearsal strategies actively address forgetting by storing and reintroducing past data examples or their generated representations.

- **Experience Replay (ER)** [15] is a rehearsal technique where a small episodic memory buffer is maintained containing a sample of old, representative examples (images and labels). When training on a new task, mini-batches are constructed by mixing data from

the new task with a selection of examples from this buffer. This concurrent training helps the model retain a base level of performance on previously learned classes. While it serves as a strong baseline, its main drawbacks are the memory consumption for storing data and potential privacy concerns.

- **Average Gradient Episodic Memory (A-GEM)** [16] improves efficiency by using the buffer to constrain the direction of parameter updates rather than for full data replay. When learning a new task, a reference gradient (g_{ref}) is computed using examples from the small memory buffer. If the new task's current gradient (g) would increase the loss for previous tasks (i.e., $g_{ref} \cdot g < 0$), A-GEM projects the current gradient onto a subspace to ensure the update does not harm the performance on old tasks. This technique effectively balances a minimal memory cost with good stability against catastrophic forgetting.

- **Dark Experience Replay (DER)** [17] enhances ER by incorporating knowledge distillation on the replayed examples. In addition to storing images and labels, DER saves the original model's outputs, or logits (the "dark experience"), for the buffer examples. When training on a new task, the overall loss includes a Distillation Loss that forces the current model's output for the replayed buffer examples to remain close to the saved old logits, thereby maintaining consistency and stability. Dark Experience Replay ++ (DER++) is an evolution of DER that adds a standard Cross-Entropy (CE) Loss using the true labels of the replayed examples alongside the distillation loss.

  xxxxADD DER LOSS EQUATIONSxxx

- **Generative Feature Replay [18]** is a non-data-storing method that tackles both forgetting and class imbalance in CIL. For each task, the model is split into a feature extractor (F_t) and a classifier (H_t). Instead of replaying raw images, a generator (G_{t-1}) is trained to produce synthetic features of previously learned classes. When a new task arrives, the new classifier is trained using real features from the new classes and the generated synthetic features from the old classes, creating a balanced training set. The feature extractor's stability is maintained using a feature distillation loss applied against the old extractor (F_{t-1}). This method leverages the fact that generating features is significantly easier and more privacy-preserving than generating complex images.

  xxxADD Figure 1 from Generative feature paperxxx

2.3.5 Regularization Methods

- Elastic Weight Consolidation (EWC) [19]
- Synaptic Intelligence (SI) [20].
- Learning Without Forgetting (LwF) [21].
- Adaptive Plasticity Improvement (API) [4].
- Inference Free Low-Rank Adaptation (InFLoRA) [22].
- Distributed Online LoRA for Federated INcremental learning method (DOLFIN) xxxADD WHEN RELEASEDxxx.

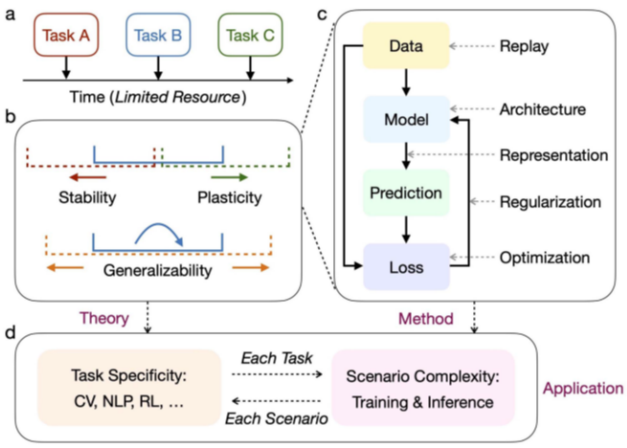


Figure 2.2: Four key aspects of CL framework [11].

2.3.6 Representation Methods

- Learning to Prompt (L2P) [23].
- Dual Prompt [24].
- CODA-Prompt [25].
- Hierarchical Generative Prototypes (HGP) [26].

In summary, as illustrated in Figure 2.2, the conceptual framework of continual learning can be understood through four interrelated components. First, continual learning involves adapting to incremental tasks under dynamic data distributions, reflecting the non-stationary characteristics of real-world environments. Second, an effective continual learning system must strike an appropriate balance between stability and plasticity: stability ensures the retention of previously acquired knowledge, whereas plasticity enables the integration of novel information. Moreover, the model should generalize effectively to both intra-task variations, which represent differences within a single task, and inter-task variations, which capture distributional shifts across tasks. Third, to achieve these objectives, a variety of methodological approaches have been proposed—each addressing different facets of the learning process. Unlike those presented in [11], the state-of-the-art methods analyzed in this work are categorized into rehearsal, regularization, and representation-based approaches. Finally, continual learning techniques must also be tailored to practical scenarios, where factors such as task specificity, scenario complexity, and computational limitations further influence the design and evaluation of effective solutions [11].

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